



香港大學

THE UNIVERSITY OF HONG KONG

CIVL 7023 Project 1

3D Quality Inspection of Prefabricated Components

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Project 1: today's focus

A measurement framework for prefabricated components

The project brief

Review methods and identify their drawbacks.
Design a task-aware, dual-head learning framework.
Validate it against comparison methods.

What you should understand today

How a scan becomes a labelled point cloud.
How geometry becomes a measured quantity.
How a measured quantity supports a quality assurance decision.

Project's route

Quality need LiDAR Scanning 3D understanding Measurement Evaluation

Why geometric quality matters



Prefabricated components



Transport and assembly

A panel that is too wide, an opening in the wrong place, or a misplaced rebar can prevent the next connection from fitting.

What do we inspect?



Inspection target	Typical question
Overall dimensions	Are length, width and height correct?
Openings and interfaces	Are size, location and spacing correct?
Wall surfaces	Are surface deviations acceptable?
Protruding rebars	Are length and position correct?

Geometric inspection checks shape and position. Material strength and hidden defects need other tests.

Manual inspection: the reference workflow



1. Establish a reference

Use drawing axes, a datum and suitable measuring tools.

2. Measure selected locations

Read a tape, straightedge, level or other calibrated tool.

3. Compare and record

Check the specified tolerance and record the result.

The problems a 3D workflow must solve



Limited sampling

A few readings can miss a local bulge or an incorrectly placed feature.

Repeated work

Each component requires access, setup, measurements and record keeping.

Difficult scenes

Occlusion, clutter and rough surfaces also challenge automated inspection.

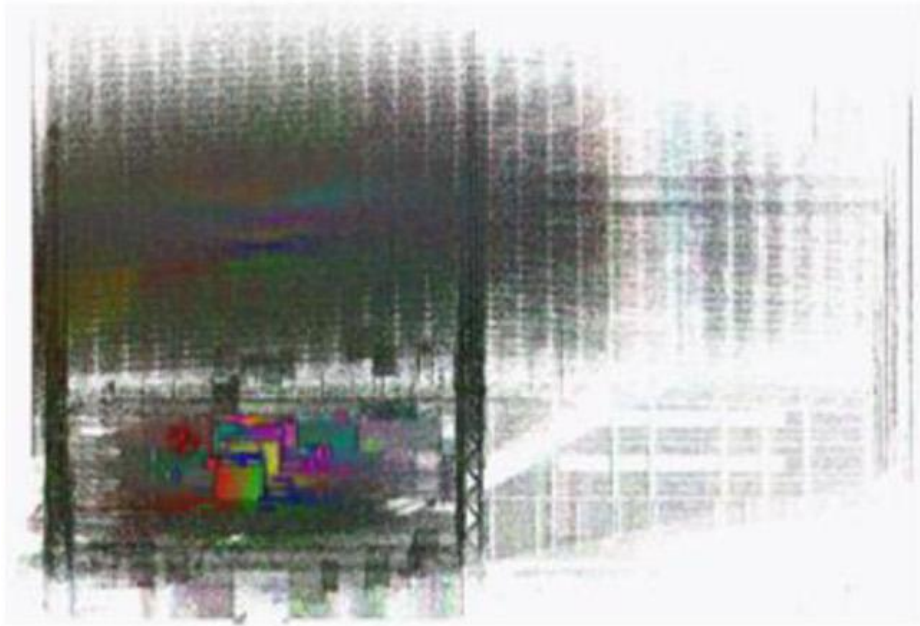
The scan-to-inspection workflow

1	2	3	4	5
Capture	Prepare	Understand	Measure	Compare
Raw scan points	Clean, aligned cloud	Objects and point labels	Dimensions and deviations	Tolerance decision

As-designed CAD / BIM supplies geometry, reference datums and required dimensions.

An inspection specification supplies the measurement method and allowable tolerance.

Point clouds: surfaces represented by points



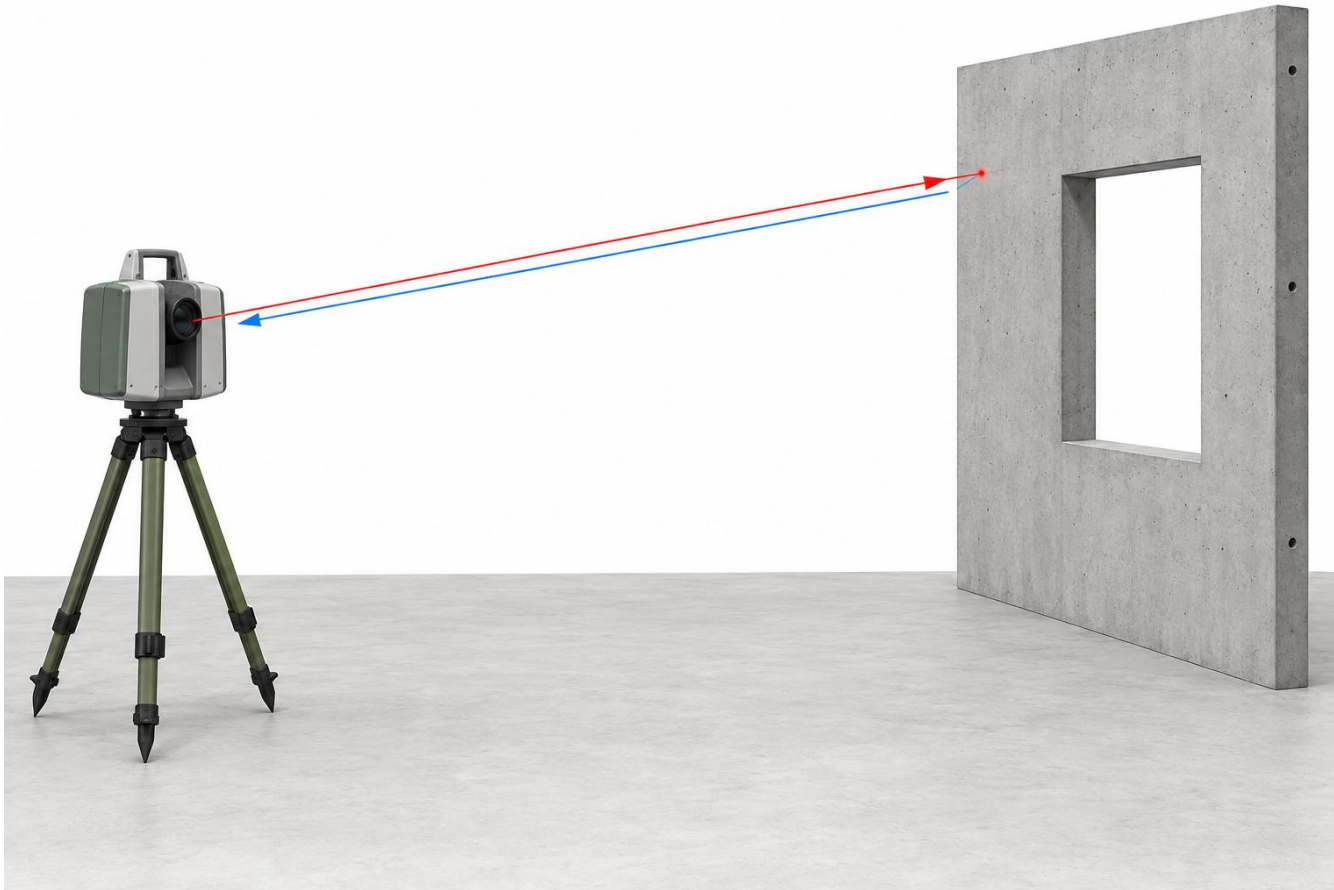
Each point stores a 3D position: (x, y, z) .

Point	x (m)	y (m)	z (m)
P1	1.250	0.005	0.800
P2	1.260	0.006	0.800
P3	1.270	0.004	0.810

Illustrative coordinates, not experimental data

Optional attributes: reflected intensity, RGB colour and time.
A point cloud has samples and gaps, rather than continuous surfaces.

How LiDAR measures a surface point



Time of flight

$$r = c \times \Delta t / 2$$

r : distance to the surface

c : speed of light

Δt : round-trip travel time

Distance + two beam angles
give one (x, y, z) point.

A changing beam direction samples many points. Some scanners use phase-based ranging.

Step 1: choosing a scanning system

System and examples	Main advantage	Main limitation for inspection
Static terrestrial laser scanner Leica RTC360, Trimble X9, FARO Focus	Stable scans of components Detailed surface coverage	Several stations and registration Occluded faces still need access
Mobile SLAM system NavVis VLX 3, Leica BLK2GO	Rapid capture while walking Useful for factory or site context	Trajectory drift and local detail Need independent dimensional checks
Robot LiDAR sensor Livox MID-360, Ouster OS series	Continuous capture Flexible integration with robots	Calibration and motion compensation Sensor precision may be insufficient

For millimetre-level component checks, validate the complete measurement workflow against the required tolerance.

Representative terrestrial laser scanners



Trimble X9

Product family	Workflow emphasis
Leica RTC360	Fast capture and VIS-assisted registration between stations
Trimble X9	Automatic calibration and levelling In-field registration in Perspective
FARO Focus	Range configurations and SCENE / Stream workflows

Compare 3D point accuracy at the same range and surface, point spacing, registration checks, software and total effort.

Scan planning and visible surfaces



Coverage

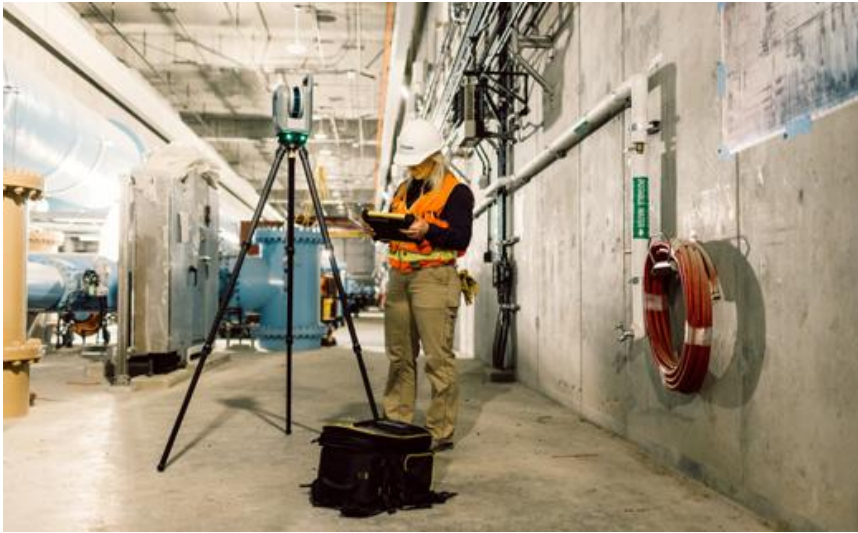
Scan critical faces and interfaces from several suitable positions.

Overlap

Neighbouring scans need shared geometry or targets for alignment.

**A hidden surface gives no direct measurement.
Reposition the scanner or record “not inspected”.**

Field acquisition: an operational checklist



**Output: raw scans, station IDs,
control observations and scan log**

1. Define the inspection

Select the component, features, drawing revision and tolerance.

2. Set up the capture

Stabilise the object. Place stations to see edges, faces and tips.

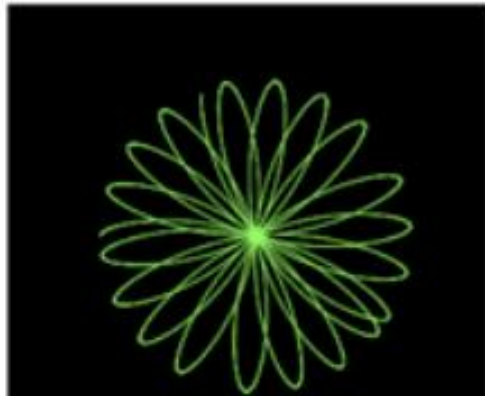
3. Check the data on site

Inspect overlap and missing areas. Scan a reference length or check targets.

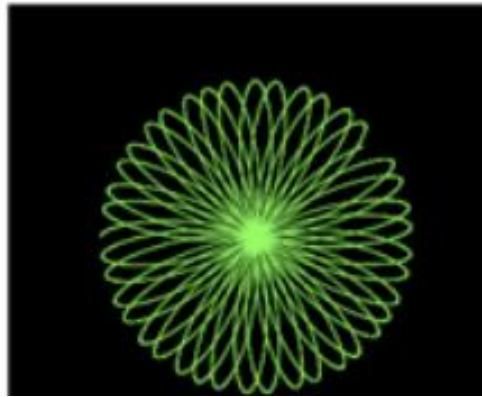
4. Export with traceability

Keep native files. Export E57 or another supported format with units and coordinates.

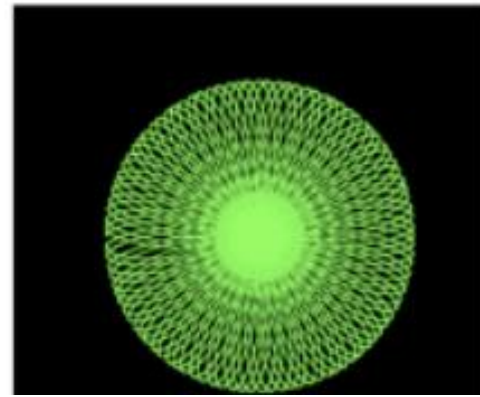
Point density and measurement accuracy



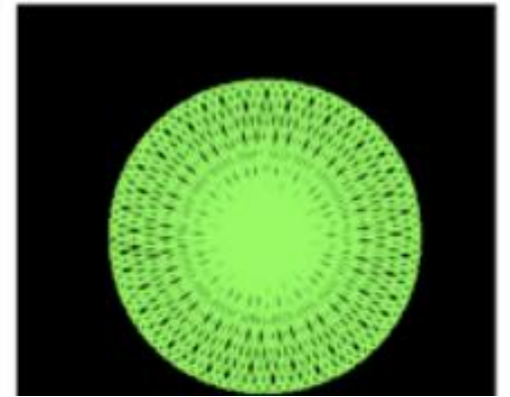
0.1 s



0.2 s



0.5 s

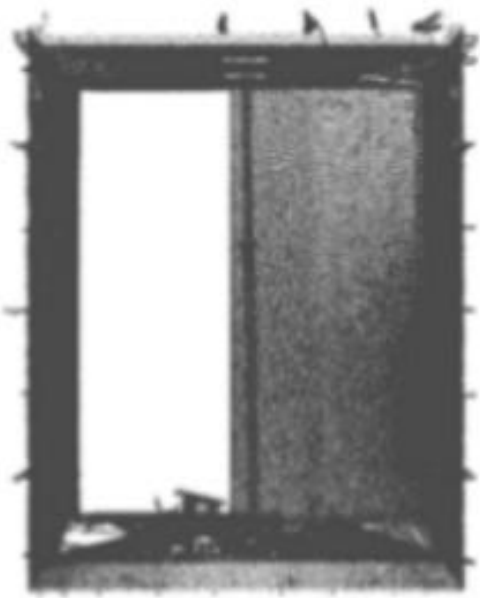


1.0 s

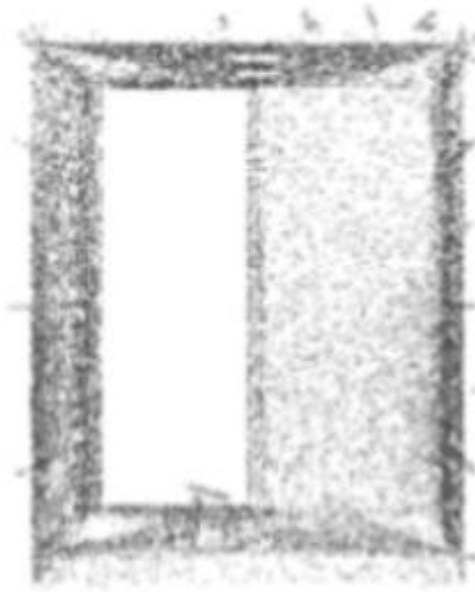
Longer accumulation fills more of this scanner's sampling pattern.

Point spacing controls visible detail. Range noise, incidence angle and alignment also affect the measurement.

Preparing a point cloud for analysis



Dense scan



Reduced sample

Crop and filter

Keep the inspection region and remove invalid or isolated points.

Voxel downsampling

Divide space into small cells and retain a representative point per cell.

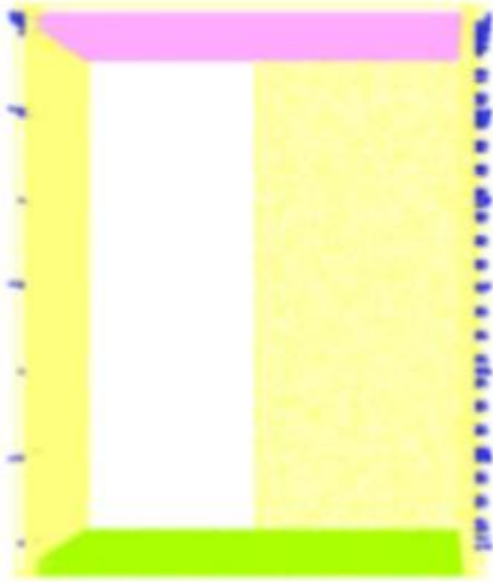
Preserve the original cloud for final measurement. Check that filtering has not removed a real defect.

Step 2: common point-cloud processing methods

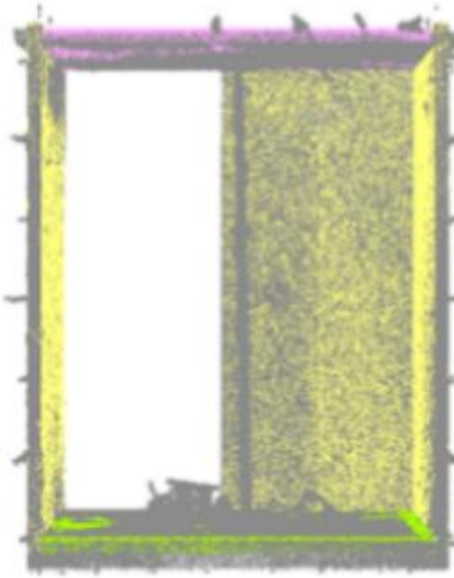
Method	What it does	What to check
Crop / pass-through filter	Keeps the inspection region	Do not cut away a required edge or tip
Statistical outlier removal	Flags unusually isolated points	Sparse valid features can look isolated
Voxel downsampling	Reduces point count using spatial cells	Cell size must preserve the smallest feature
Normal estimation	Estimates local surface directions	Neighbourhood size changes the result
RANSAC / region growing	Extracts model or smooth-surface groups	A geometric group still needs a semantic label

Save processing parameters and point IDs. Transfer labels to the original metric cloud for final measurement.

Registration: a common coordinate frame



Design reference



Aligned geometry

Two alignment tasks

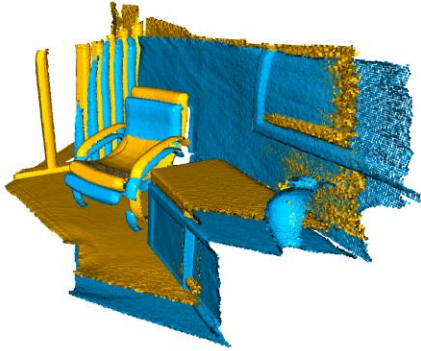
Combine scans from different stations.
Align the measured component with
CAD / BIM.

Coarse alignment, then ICP

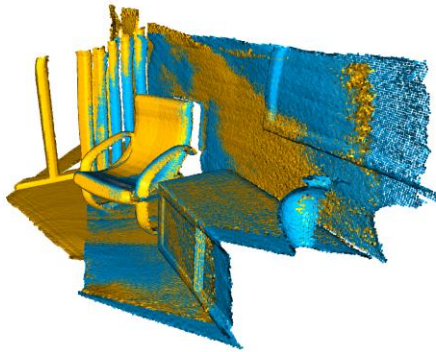
Start with targets or matching features.
Iteratively refine point
correspondences.

Use rigid rotation and translation. Preserve real dimensional deviations.

ICP registration: what the iterations do



Initial alignment



Refined alignment

1. Start close

Use targets, corresponding features or a controlled datum.

2. Match nearby points

Find plausible correspondences in overlapping regions.

3. Estimate a rigid update

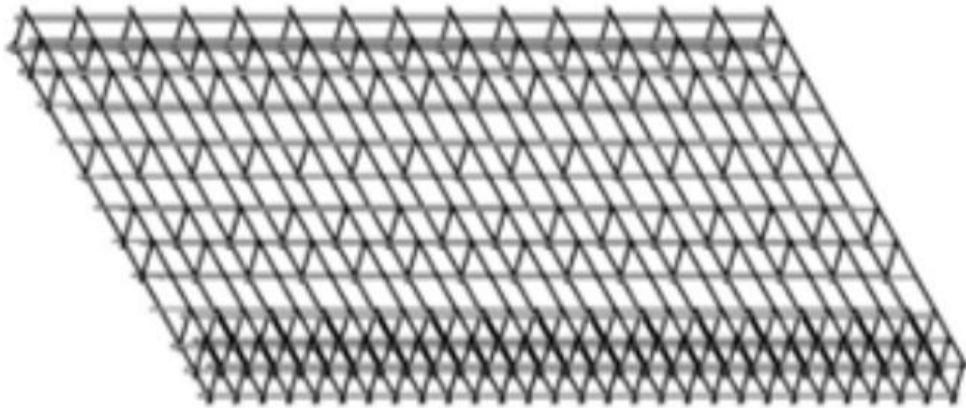
Minimise point-to-point or point-to-plane residuals.

4. Repeat and verify

Update matches until stable. Check independent targets and critical faces.

Keep scale fixed. A small registration residual alone does not prove correct inspection alignment.

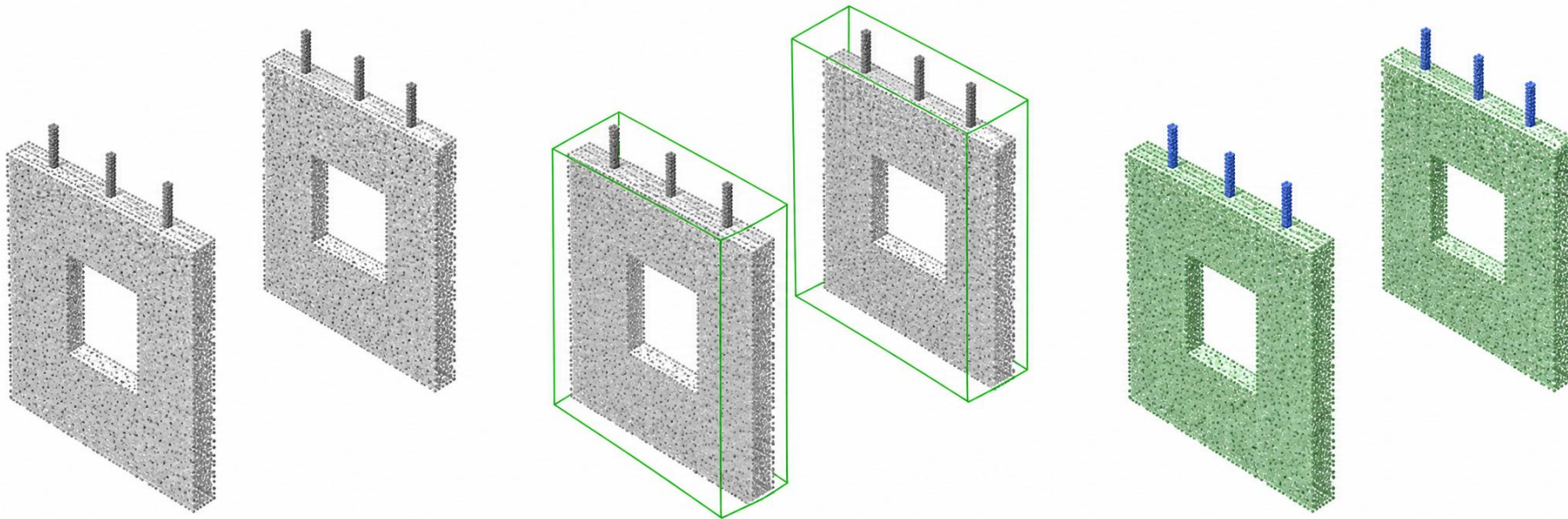
Design values, datums and tolerances



Required input	Example for a wall panel
Nominal geometry	Width = 3,000 mm
Datum	Defined base plane and side reference
Measurement definition	Distance between specified side faces
Tolerance	An agreed allowable interval

Example values are for teaching. Use the applicable drawing and approved inspection specification in a real project.

Detection and segmentation in 3D



Raw point cloud

Measured points,
without object labels

Detection

Where is each object?
A box, class and score

Segmentation

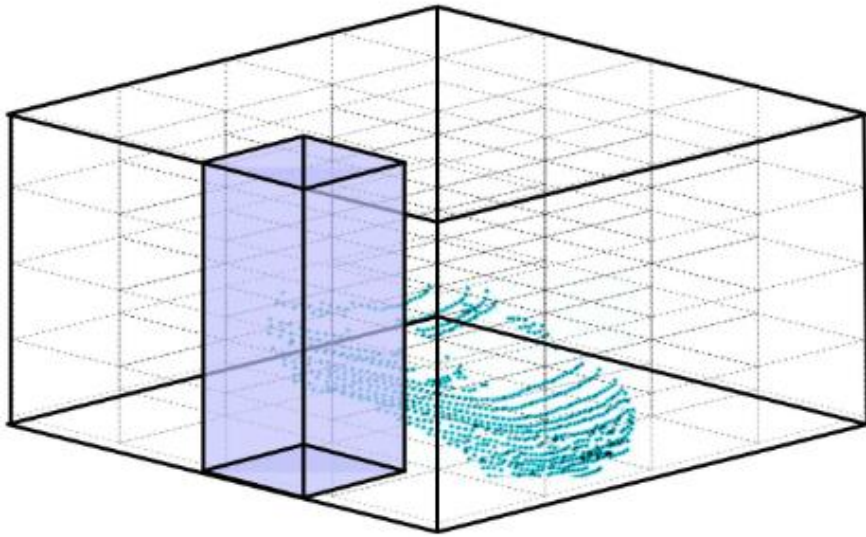
Which class is each point?
Concrete or rebar, for example

Step 3: methods for automatic semantics

Method	How it assigns meaning	Good fit / main limitation
Geometry rules and BIM	Planes, normals, size and registered design objects	Simple repeated components Rules depend on layout and alignment
Features + classifier	Local shape / colour features with Random Forest or SVM	Modest labelled datasets Handcrafted features need tuning
Deep 3D models	PointNet++, sparse 3D CNNs or point transformers	Complex shape and context Need representative training data
Image / foundation-model transfer	Use calibrated images, masks and 3D projection or fusion	Useful semantic proposals Occlusion and fine boundaries need review

Clustering groups points. Semantic classification gives those groups or points a meaning.

How a deep 3D model learns geometry



Point-based models

PointNet learns from point sets. PointNet++ also learns local neighbourhood geometry.

Voxel-based models

A 3D grid organises the points for spatial feature learning.

**Learned features support detection or point labels.
The model still needs representative training examples.**

Training and inference

Training	Inference on a new scan
Scans with labelled components and features	A scan without known answers
Predict labels and compare with annotations	Apply the trained model
Update model parameters to reduce errors	Return predictions and confidence scores
Validate on separate components or sites	Check unfamiliar or uncertain cases

Split data by physical component or site. Nearby points from the same scan are not an independent test.

Semantic predictions into measurable objects

1. Prepare model input

Use the trained feature format and normalisation. Retain original XYZ and point IDs.

2. Predict point labels

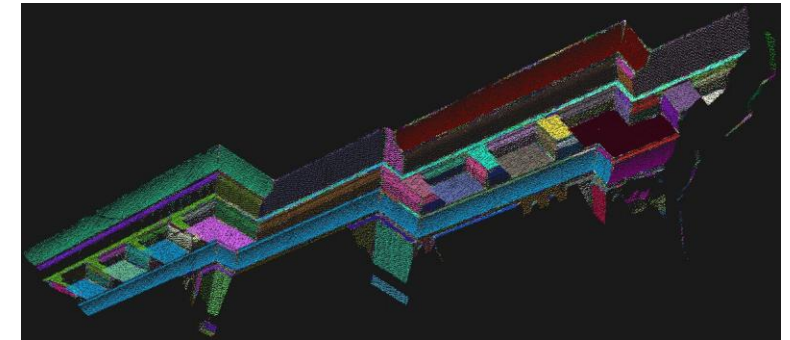
Assign wall, rebar or background probabilities. Keep uncertain points flagged.

3. Separate individual objects

Within a class, use connected components, DBSCAN or an instance model.

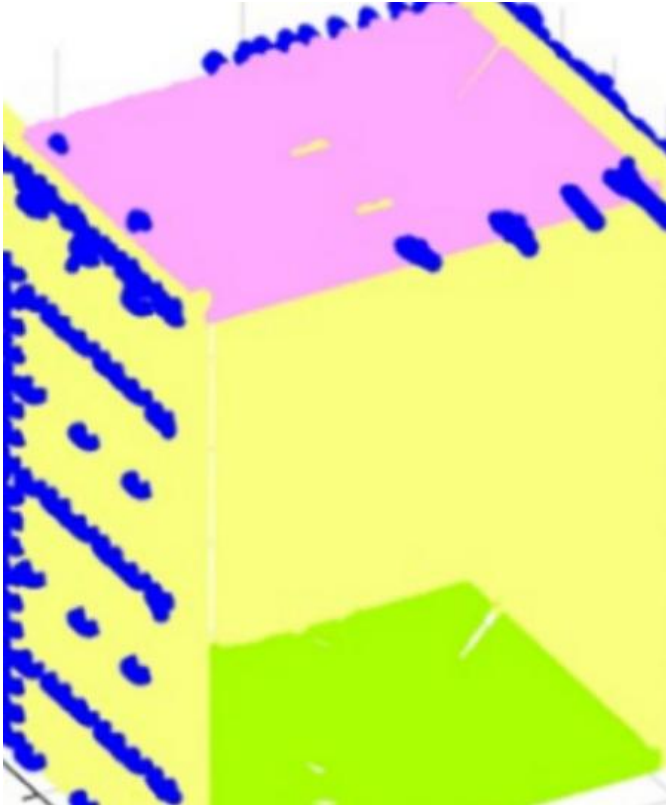
4. Recover measurement points

Map results to original coordinates. Check the boundaries, tips and visibility.



Geometric groups
are a useful starting point.

Labels must preserve measurement features



The right category

Separate wall, ceiling, floor, rebars and temporary objects.

The right boundary

Keep the edge or tip that defines the quantity being measured.

A small boundary error can change the result even when most point labels are correct.

Research example: BIM-guided segmentation

Geometric fitting: a stable reference surface

Plane model: $ax + by + cz + d = 0$



A wall cross-section contains surface points and unwanted observations.

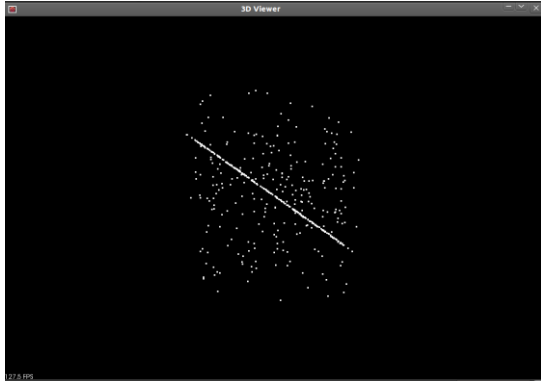
RANSAC

Propose a plane from a small sample. Count supporting points. Repeat and refine.

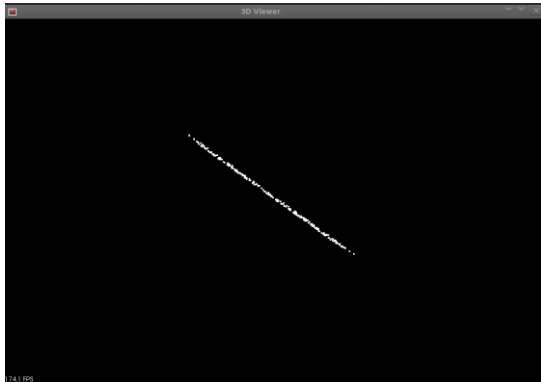
A measurement safeguard

Inspect rejected points. A real protrusion must not disappear as “noise”.

RANSAC plane fitting: six steps



Points containing outliers



Supporting plane points

1. Set a distance threshold

Choose ϵ using noise and the feature scale.

2. Sample three points

Reject samples that lie on one straight line.

3. Fit a candidate plane

Estimate its normal and offset.

4. Count supporting points

A point is an inlier if its plane distance $\leq \epsilon$.

5. Repeat and select

Keep a well-supported, task-relevant plane.

6. Refit and inspect

Refit on inliers. Retain rejected points for review.

RANSAC threshold and quality tolerance

Illustrative point distances from a candidate plane. All values are teaching assumptions.

Point	P1	P2	P3	P4	P5	P6
Signed distance (mm)	-1	0	+1	+2	+8	+14
Inlier for $\varepsilon = 3$ mm?	Yes	Yes	Yes	Yes	No	No

Plane-fitting threshold: $\varepsilon = 3$ mm

Selects points that support a geometric model.

Quality tolerance: for example ± 5 mm

Defines allowable deviation under the inspection specification.

A rejected point at 8 mm could be a real bulge. Inspect the original surface after fitting.

Measuring dimensions and opening sizes



Example: panel width

Fit the intended left and right faces.
Express both in the component coordinate frame.
Measure their separation in the specified direction.

$$W = x(\text{right}) - x(\text{left})$$

For an opening, fit its visible boundary or reveal faces and apply the same dimensional definition.

Step 4: matching the method to the quantity

Method	Suitable measurement	Main limitation
Fitted planes / lines	Panel dimensions, opening width, spacing between intended faces	Needs visible faces and a clear direction or inspection section
Cloud-to-mesh distance	Local deviation from a CAD / BIM surface after datum alignment	Wrong alignment or mesh normals can distort interpretation
Reference-plane residuals	Surface variation about a specified or fitted reference	A fitted-plane statistic is not a universal flatness definition
Axis / cylinder + tip	Straight rebar direction, position and protrusion	Thin, bent or partly hidden bars can violate the fitted model
Cloud-to-cloud distance	Screening changes between scans	Sampling and density affect the nearest-point distance

The drawing defines the quantity. The algorithm must measure that same quantity.

Dimensional inspection: a worked procedure

1. Select the required faces

Use semantic labels or manual segmentation to isolate the two side faces.

2. Establish the datum

Transform into the component frame using trusted references.
Keep scale fixed.

3. Fit and inspect each face

Use RANSAC with refinement, or least squares on a clean selection. Check residuals.

4. Compute the specified width

Measure along the required axis at the stated section or height.

5. Compare and save evidence

Calculate the deviation, apply the decision rule and save the fitted geometry.

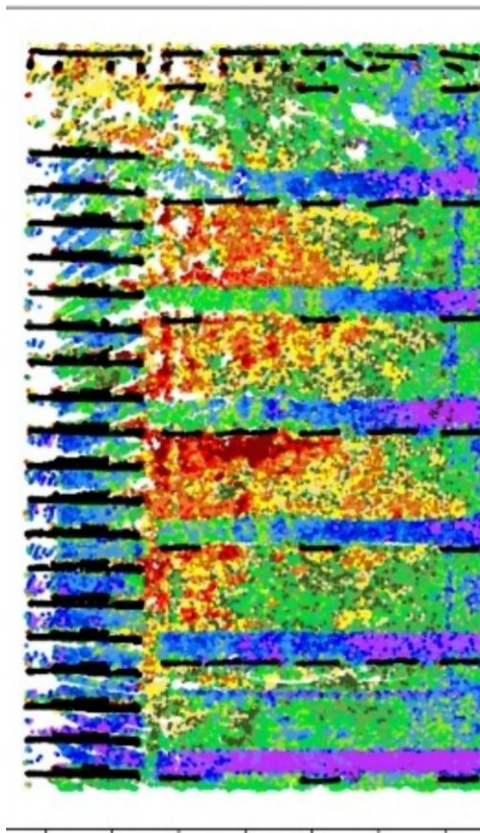
Teaching example	Value
Left face x	0 mm
Right face x	3,004 mm
Design width	3,000 mm
Measured width	3,004 mm
Deviation	+4 mm

CloudCompare example

Segment each face

Tools > Fit > Plane

Surface deviation and flatness



Surface deviation pattern only

Quantity	Reference and meaning
Design deviation	Signed distance to the specified design surface
Plane-fit variation	Variation of distances about a fitted plane
Specified flatness	Use the stated method, such as a straightedge or minimum-zone criterion

The colour map needs a distance scale, units and a stated reference before it supports a decision.

Surface inspection in CloudCompare

1. Load and align

Load the scan and CAD mesh in the same units. Use the prescribed datum alignment.

2. Choose the reference

Select the inspected cloud and the corresponding design mesh or fitted plane.

3. Calculate surface distances

Tools > Distances > Cloud/Mesh Dist. Check signed distances and normal orientation.

4. Interpret and report

Set a labelled scale in millimetres. Inspect exceedance regions and missing coverage.

Which reference?

CAD mesh

Deviation from the nominal design

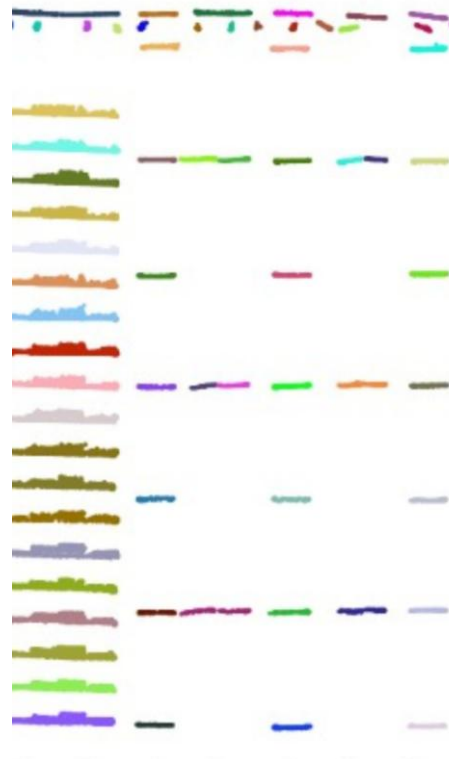
Fitted plane

Variation about an estimated surface

Specified flatness

Apply the required straightedge or minimum-zone procedure

Rebar protrusion and interface features



Each rebar needs its own instance.

Identify the reference wall surface.

Locate the visible tip.

Measure in the specified direction.

$$L = (p(\text{tip}) - p(\text{base})) \cdot u$$

u is the unit vector along the required measurement direction.

Worked example: comparison with design

Illustrative panel inspection. All values and tolerances below are teaching assumptions.

Item	Design (mm)	Measured (mm)	Deviation (mm)	Allowed	Point estimate
Panel width	3,000	3,004	+4	±5 mm	Within
Opening width	900	892	-8	±5 mm	Outside
Rebar protrusion	100	103	+3	±5 mm	Within

Deviation = measured value - design value

For a symmetric tolerance $\pm T$, the point estimate is within the interval when $|\text{deviation}| \leq T$.

Uncertainty near the tolerance limit

Classroom decision: is the panel width acceptable?

Design: 3,000 mm Tolerance: ± 5 mm

Measured: 3,004 mm Estimated uncertainty interval: ± 2 mm

Deviation interval: +2 mm to +6 mm

The interval crosses the +5 mm limit. Apply the agreed decision rule and review or remeasure the feature.

An inspection report connects result to evidence

Report field	Example record
Component and design revision	Panel P-017, drawing revision C
Inspection item and datum	Opening width at the defined inspection height
Measured result	892 mm; deviation -8 mm; tolerance ± 5 mm
Decision and action	Outside interval by point estimate; hold for review
Supporting evidence	Labelled points, fitted edges, scan views and raw file ID
Traceability	Date, operator, instrument, method and uncertainty

Unseen or unreliable features receive an explicit “not inspected” or “review” status.

Research example: BIM and LiDAR inspection

Tan et al. (2024)



Modular box and scanner in the factory

Four real modular boxes in a factory.
BIM-guided extraction, registration and segmentation.
Reference-plane measurement of wall surfaces and rebars.

Comparison with manual measurements	Pairs	RMSE
Outer wall surface inspection	80	2.6 mm
Rebar protrusion inspection	88	2.8 mm

What the case demonstrates, and what remains

Same four-box case	Elapsed inspection time
Manual: 4 × 78 min	312 min
BIM + LiDAR workflow	184 min
Reduction	128 min, approximately 41 %

Evidence of potential

The study reported less total inspection time and dense spatial results.

Remaining limitations

People still plan scans and set parameters. Occlusion, clutter and registration affect reliability.

Case-specific results. Elapsed time is not the same as labour-hours or total cost.

Manual and 3D inspection: complementary strengths

Aspect	Manual inspection	LiDAR and 3D processing
Coverage	Selected accessible locations	Dense coverage of visible surfaces
Setup and cost	Simple tools, flexible setup	Scanner, processing and validation effort
Measurement quality	Depends on tool, operator and datum	Depends on sensing and the full pipeline
Records	Readings, notes and photographs	Spatial evidence that can be revisited
Difficult cases	People interpret unusual conditions	Occlusion and unfamiliar clutter need review

A practical workflow uses 3D screening and evidence, with targeted manual verification where needed.

Project 1: a defensible experiment

A focused task

Choose the component, feature, datum and measurement definition.

Two kinds of evidence

Report segmentation quality and measurement error in millimetres, plus runtime and failure cases.

Fair comparisons

Compare a geometric baseline, a suitable deep model and your dual-head design.

An independent test

Use held-out components. Compare with trusted reference measurements and state their uncertainty.